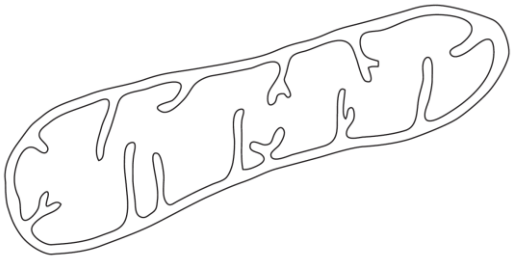
Structured questions

|!|BCELA0220000001|!| ★★ **SI**

The diagram below shows the structure of a mitochondrion.



(a) On the diagram above,

(i) label the site where oxidative phosphorylation occurs with the letter P. (1 mark)

(ii) label the site where the Krebs cycle occurs with the letter K. (1 mark)

(b) Antimycin A is an inhibitor of oxidative phosphorylation that acts by binding to one of the electron carriers. Scientists carried out an experiment to investigate the effect of Antimycin A on the respiration of yeast cells.
In the experiment, yeast cells were mixed with a buffer solution containing ADP, phosphate ions and glucose. The mixture was then incubated in a 30 °C water bath for 30 minutes. During this time, the oxygen content of the mixture was measured.
The experiment was then repeated, plus the addition of Antimycin A into the mixture five minutes after the start of incubation.
The following table shows the results.

Incubation time (min)	**Oxygen content of the mixture (arbitrary units)**	
	Without Antimycin A	**With Antimycin A**
0	6.5	6.5
5	4.0	4.0
10	2.5	4.0
15	1.7	4.0
20	1.0	4.0
25	0.5	4.0
30	0.5	4.0

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(i) Explain why the oxygen content of the mixture without Antimycin A did not decrease to zero. (2 marks)

(ii) Explain why the oxygen content of the mixture did not decrease after Antimycin A was added. (2 marks)

(iii) State and explain how the addition of Antimycin A will affect the production of ATP in yeast. (3 marks)

##

(a) (i) Label inner membrane as P (1)

(ii) Label matrix as K (1)

(b) (i) Any ***one set*** of the following: (2)

- The reactants (e.g. glucose, ADP, phosphate) have been used up, (1) so there is no source of hydrogen / electrons to react with oxygen. (1)
- Yeasts switch to anaerobic respiration in low oxygen conditions. (1) Anaerobic respiration does not involve the use of oxygen. (1)

(ii) Antimycin A prevents electrons from being passed along a series of carriers. (1)

Thus oxygen was not reduced and cannot react with hydrogen ions to form water molecules. (1)

(iii) ATP production decreases. (1)

Antimycin A inhibits oxidative phosphorylation, which is the process that produces the most ATP. (1)

The Krebs cycle stops because no NAD / FAD (which is essential for the Krebs cycle to occur) is regenerated from oxidative phosphorylation. (1)

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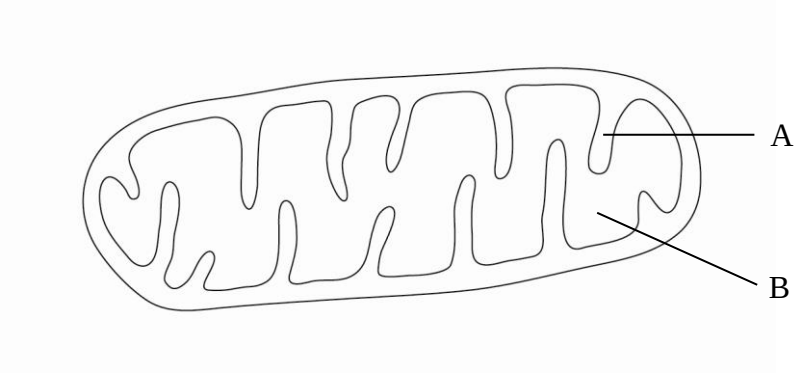
(9 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	2
(b)(iii)	Applying concepts	3

|!|BCELB0220000002|!| ★

The diagram below shows a mitochondrion.



(a) Which stages of respiration occur in structures A and B, respectively? (2 marks)

(b) Each mitochondrion contains a large number of structure A. Explain the advantage of this. (2 marks)

(c) People suffering from mitochondrial disease have mitochondria that do not function properly and they can only exercise for short periods of time.

(i) Explain why people with mitochondrial disease can only exercise for a short time. (2 marks)

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(ii) Explain why people with mitochondrial disease often have a high level of pyruvate in their blood after a meal. (2 marks)

##

(a) A: oxidative phosphorylation (1)
B: the Krebs cycle (1)

(b) Structure A increases the surface area for the attachment of enzymes and carrier molecules, (1)
which facilitates the production of ATP. (1)

(c) (i) Muscle contractions require energy / ATP. (1)
Dysfunctional mitochondria cannot release sufficient energy / produce sufficient ATP through aerobic respiration. (1)

(ii) Glucose from meals is broken down into pyruvate molecules by glycolysis, which occurs in the cytoplasm. (1)
These pyruvate molecules cannot be metabolized by non-functional mitochondria and accumulates in the blood. (1)

(8 marks)

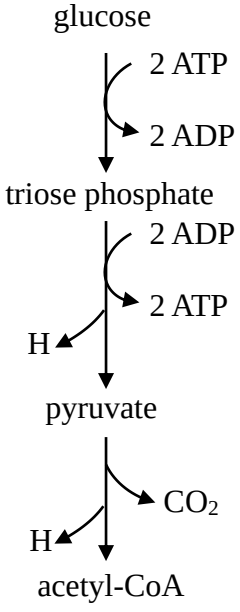
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	2
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	2

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|!|BCELB0220000003|!| ★

The following figure shows a simplified pathway of the early stages of respiration.



(a) (i) Name the process in which glucose is converted to pyruvate. (1 mark)

_____

(ii) State the location where the process in (a)(i) occurs in a cell. (1 mark)

_____

(b) Explain why ATP is needed at the start of respiration. (1 mark)

_____

_____

(c) Describe what happens to the hydrogen released in the early stages of respiration. (2 marks)

_____

_____

_____

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(d) Red blood cells lack mitochondria.

(i) Suggest how the pyruvate formed in the early stages of respiration is further metabolized in red blood cells. Explain the significance of this metabolism in sustaining the process in (a)(i). (2 marks)

(ii) Explain how the lack of mitochondria facilitates the function of red blood cells. (3 marks)

##

(a) (i) Glycolysis (1)

(ii) Cytoplasm (1)

(b) It is used to activate glucose / provide activation energy. (1)

(c) The hydrogen is accepted by NAD to form NADH. (1)

NADH then transfers the hydrogen to the inner membrane of mitochondria to carry out oxidative phosphorylation. (1)

(d) (i) Pyruvate will be reduced by NADH to form lactic acid. (1)

This will regenerate NAD for sustaining glycolysis. (1)

(ii) Red blood cells are responsible for oxygen transport. (1)

The red blood cells' lack of mitochondria means that the Krebs cycle and oxidative phosphorylation do not occur in these cells. (1)

Thus the cells do not consume any of the oxygen that it transports. (1)

(10 marks)

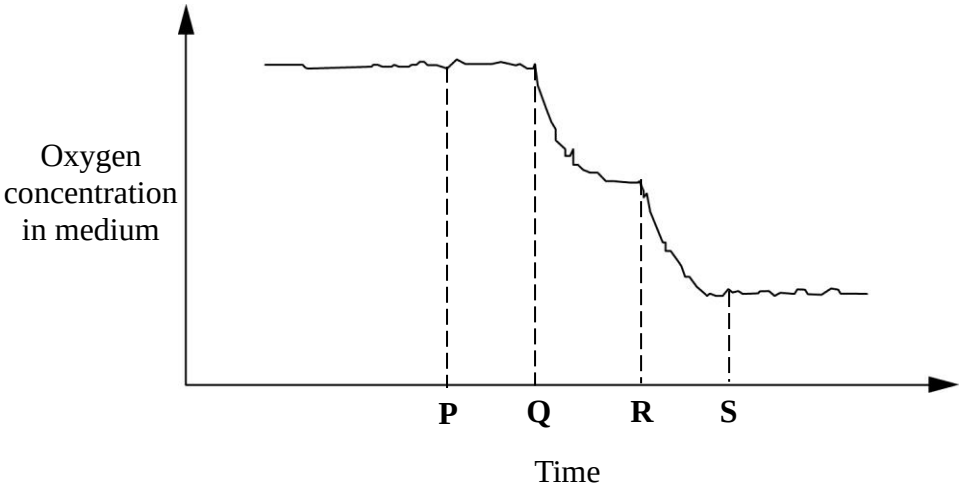
##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2
(d)(i)	Applying concepts	2
(d)(ii)	Applying concepts	3

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|!|BCELA0220000004|!| ★★ **SI**

In an investigation, liver cells were used as a source of mitochondria. These cells were first broken down to isolate the mitochondria. An oxygen sensor was used to measure the oxygen concentration in the suspended mitochondria over a period of time. At time P, the respiratory substrate glucose was added. At times Q, R and S, equal amounts of ADP were added. The following graph shows the results.



Describe and explain the results in the following time periods.

(a) between times P and Q (3 marks)

(b) between times Q and R (3 marks)

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(c) after time S (3 marks)

##

(a) The concentration of oxygen remained constant. (1)
This is because the lack of ADP was limiting the formation of ATP. (1)
Thus oxidative phosphorylation could not occur and no oxygen was consumed. (1)

(b) The concentration of oxygen decreased sharply at first, then the decrease gradually slowed and eventually ceased. (1)
The added ADP allowed oxidative phosphorylation to occur, during which ADP was converted to ATP and oxygen was consumed. (1)
The consumption of oxygen slowed down as ADP was used up. (1)

(c) The concentration of oxygen remained at a low level. (1)
This is because the substrate / glucose was used up. (1)
Thus oxidative phosphorylation could not occur and no oxygen was consumed. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)	Analysing data	1
	Applying concepts	2
(c)	Analysing data	1
	Applying concepts	2

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|!|BCELA0220000005|!| ★★ **SI**

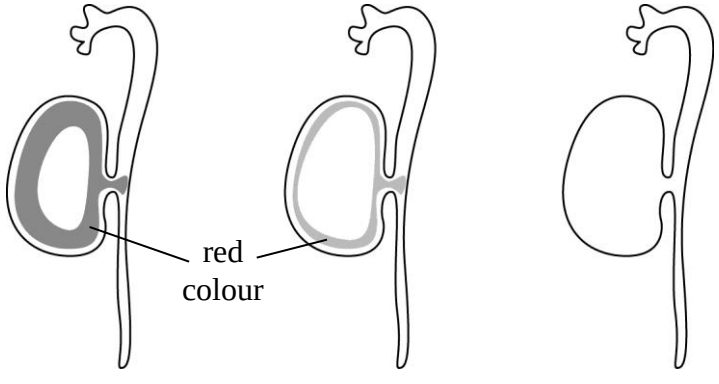
Triphenyl tetrazolium chloride (TTC) is a hydrogen acceptor. It will diffuse into respiring cells and can be reduced by hydrogen carriers (e.g. NADH) at the inner membrane of mitochondria. When TTC is reduced, it changes from being colourless to red.

Oxidized TTC ⇌ Reduced TTC
(colourless) (red)

Paul carried out an investigation into the respiratory activity of plant tissues. He used three groups of germinating seeds of the same species. The groups of seeds were first treated as shown in the following table before the investigation.

Group	***Treatment***
X	placed at 25 °C for 24 hours
Y	placed at 5 °C for 24 hours
Z	placed at 5 °C for 24 hours and then placed at 90 °C for 5 minutes

In the investigation, the seeds were cut longitudinally into two halves and placed (with the cut surface facing down) on a Petri dish with TTC solution.
The following diagrams show the appearance of the seeds after 10 minutes.



Group X **Group Y** **Group Z**

(a) Compare the appearances between the seeds of Groups X, Y and Z. (2 marks)

(b) Suggest an explanation for the results observed in the seeds of Group X. (3 marks)

(c) Suggest reasons to explain why the seeds of Groups Y and Z appeared different from those of Group X. (2 marks)

(d) When oxygen is in short supply, the germinating seeds can obtain energy from anaerobic respiration.

(i) State the products of this type of respiration. (1 mark)

(ii) Explain why respiration is essential to the growth of germinating seeds. (2 marks)

##

(a) The cotyledons of the seeds in Group X showed a greater extent of red staining than those of Group Y. (1)

The seeds in Group Z showed no red staining in the cotyledon. (1)

(b) 25 °C is a suitable temperature for respiration. (1)

More NADH was produced per unit time through glycolysis / the link reaction / the Krebs cycle. (1)

Thus more TTC was reduced by NADH to form the red stain. (1)

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(c) For Group Y:

The low temperature inactivated the enzymes involved in respiration. **OR**

The low temperature reduced the kinetic energy of the substrates and enzymes involved in respiration, thus lowering the rate at which enzyme-substrate complexes form. (1)

For Group Z:

The enzymes involved in respiration were denatured at high temperature. (1)

(d) (i) ethanol, carbon dioxide, ATP (1)

(ii) Any ***two*** of the following: (2)

- It provides energy for mitosis / cell division.
- It provides energy for protein synthesis.
- It provides energy for DNA replication.

(Or other reasonable answers)

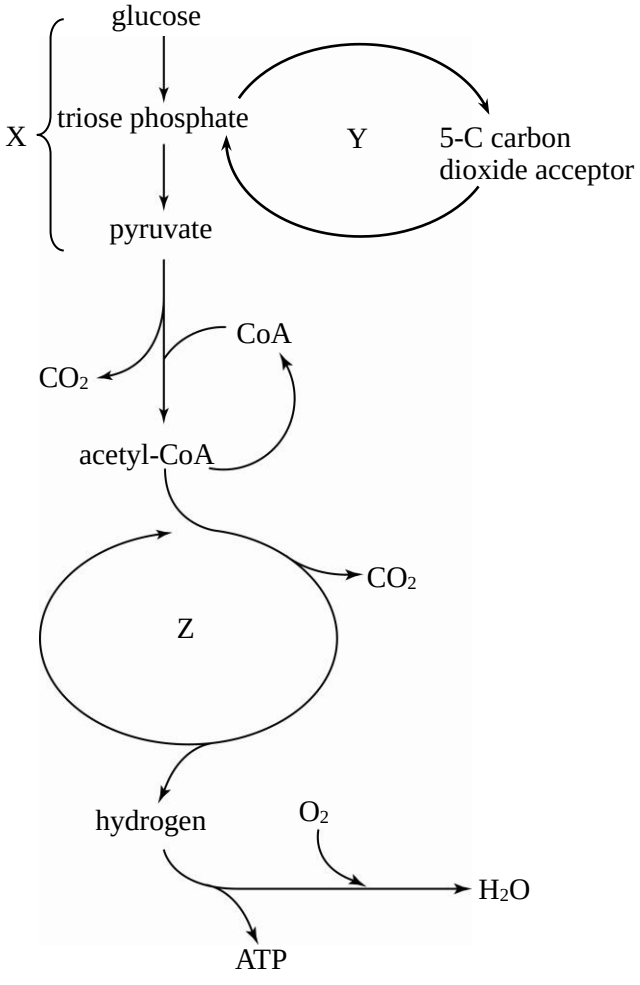
(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	3
(c)	Applying concepts	2
(d)(i)	Understanding basic concepts	1
(d)(ii)	Understanding basic concepts	2

|!|BCELB0220000006|!| ★ **Cross topic**

The diagram below shows some of the reactions that take place in a leaf cell.



(a) Name reaction pathways X, Y and Z. (3 marks)

(b) Explain how the three reaction pathways (X, Y and Z) are able to occur separately in the same leaf cell. (4 marks)

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(c) Explain the role of NAD and NADP in the leaf cell. (3 marks)

##

(a) Pathway X: glycolysis (1)
Pathway Y: the Calvin cycle (1)
Pathway Z: the Krebs cycle (1)

(b) The three pathways are separated by membranes / take place at different locations in the cell. (1)
Pathway X / Glycolysis occurs in the cytoplasm. (1)
Pathway Y / the Calvin cycle occurs in chloroplasts. (1)
Pathway Z / the Krebs cycle occurs in mitochondria. (1)

(c) NAD and NADP can accept hydrogen to form NADH and NADPH, respectively. (1)
NADH can transfer hydrogen to the carriers in the inner membrane of mitochondria for oxidative phosphorylation to occur. (1)
NADPH can supply hydrogen to the Calvin cycle. (1)

(10 marks)

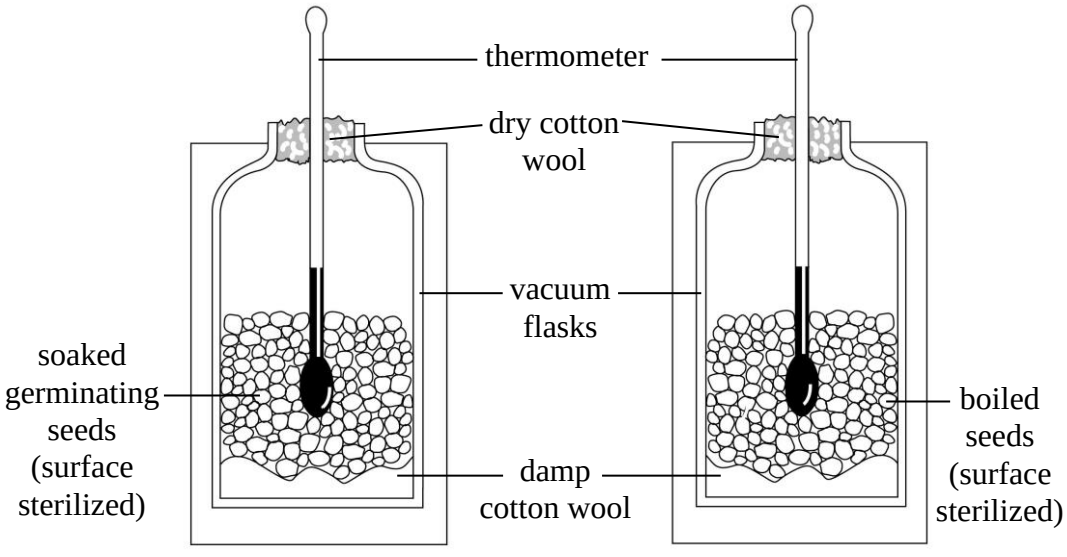
##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Applying concepts	4
(c)	Applying concepts	3

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|!|BCELB0220000007|!| ★ **SI**

The following diagram shows a set-up used to study the respiration of germinating seeds.



(a) Why were vacuum flasks used instead of plastic bottles? (1 mark)

_____

_____

(b) Why were dry cotton wool used instead of rubber stoppers? (1 mark)

_____

_____

(c) Sketch a graph to show the temperature change over time in set-ups A and B. (4 marks)

(d) State and explain the result of set-up A if the seeds were not sterilized. (2 marks)

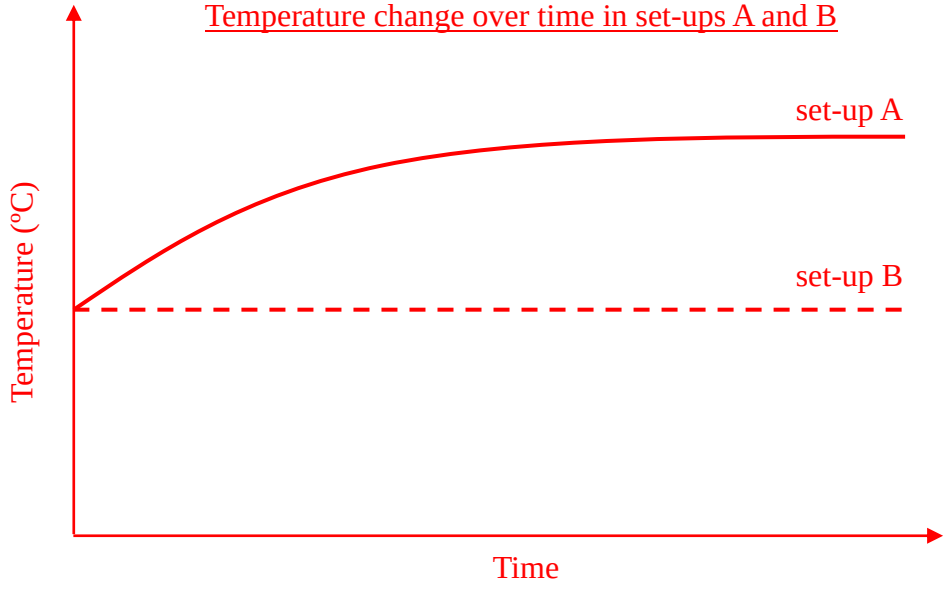
(e) Suggest and explain ***one*** improvement for the set-up. (2 marks)

##

(a) A vacuum flask is a better heat insulator that can minimize heat loss. (1)

(b) The cotton wool permitted oxygen to diffuse into the flasks, thus allowing the respiration of the germinating seeds. (1)

(c) Correct title (1)
Correct labelling of axes (1)
Correct sketching of lines (1, 1)
e.g.



(d) The temperature increase would be greater. (1)
Microorganisms on the surface of the germinating seeds would also respire and release heat, (1)
thus causing a greater temperature increase.

(e) The vacuum flasks should be inverted (1)
because warm air will rise and escape from an upright vacuum flask. (1)

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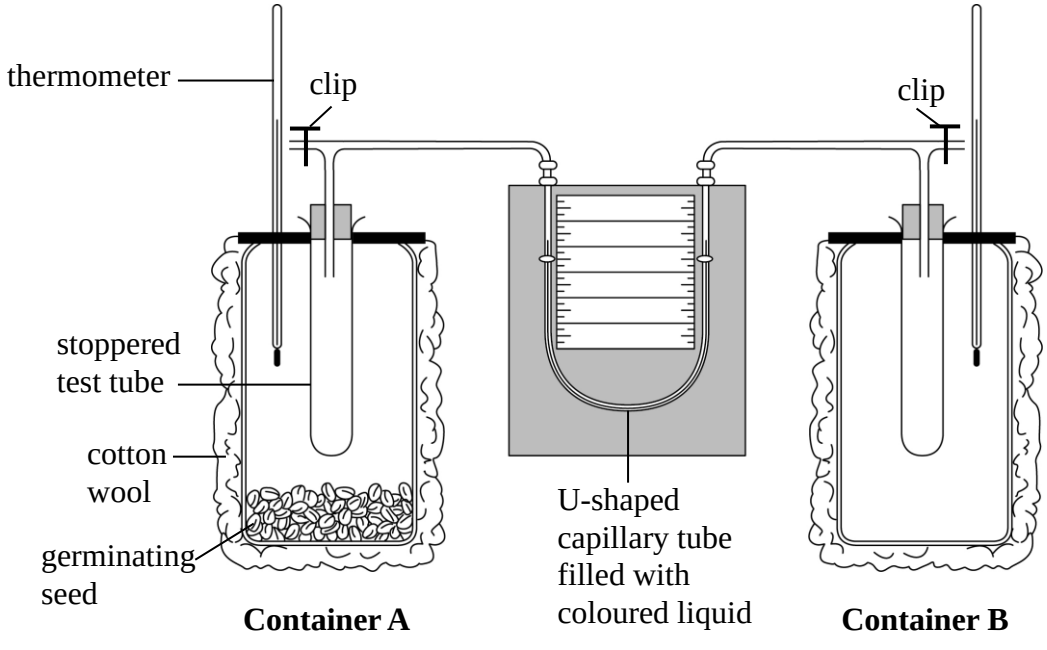
(10 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Practical skills	1
(c)	Presenting information	4
(d)	Applying concepts	2
(e)	Practical skills	2

|!|BCELC0220000008|!| **SI**

The following diagram shows the set-up used to investigate whether germinating seeds would release heat.

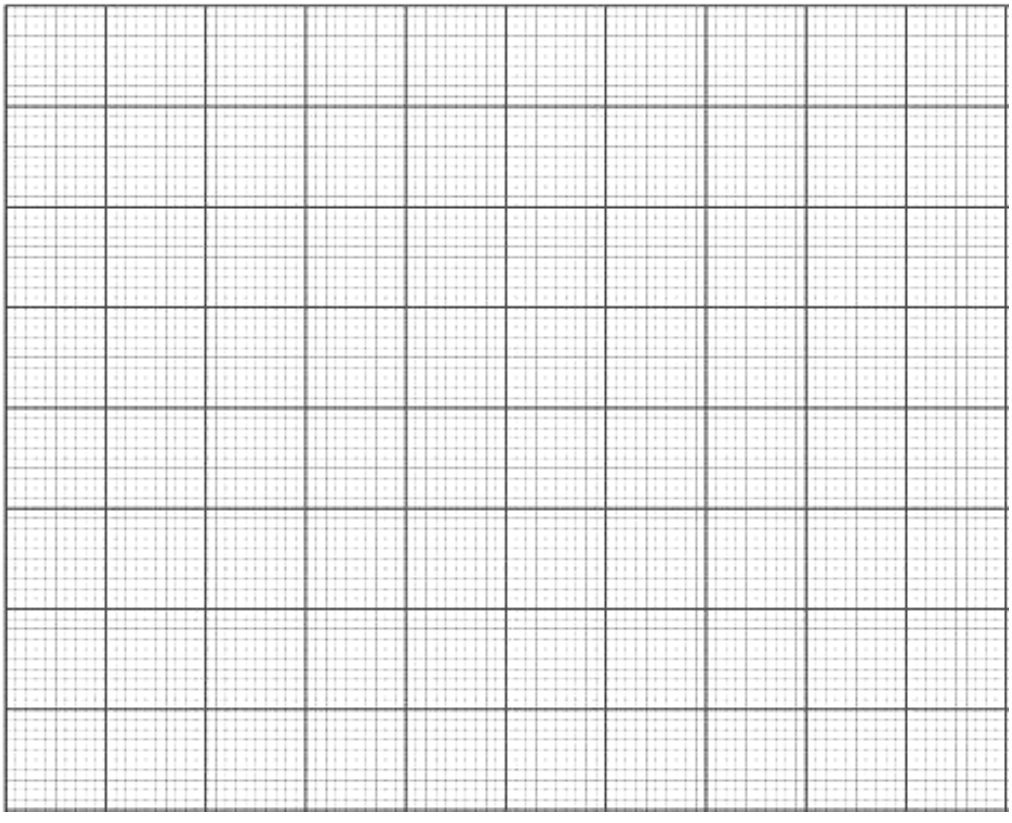


The following table shows the temperature changes in containers A and B over time.

Time (hour)	0	1	2	3	4	5	6	7	8
Temperature in Container A (ºC)	20.0	20.2	20.5	21.0	21.5	22.4	23.5	24.5	26.0
Temperature in Container B (ºC)	20.0	20.0	20.0	20.1	20.1	20.2	20.3	20.3	20.3

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(a) Plot a graph to show the temperature change in container A over time. (3 marks)



(b) Explain how the levels of the coloured liquid in the U-shaped capillary tube would change during this investigation. (3 marks)

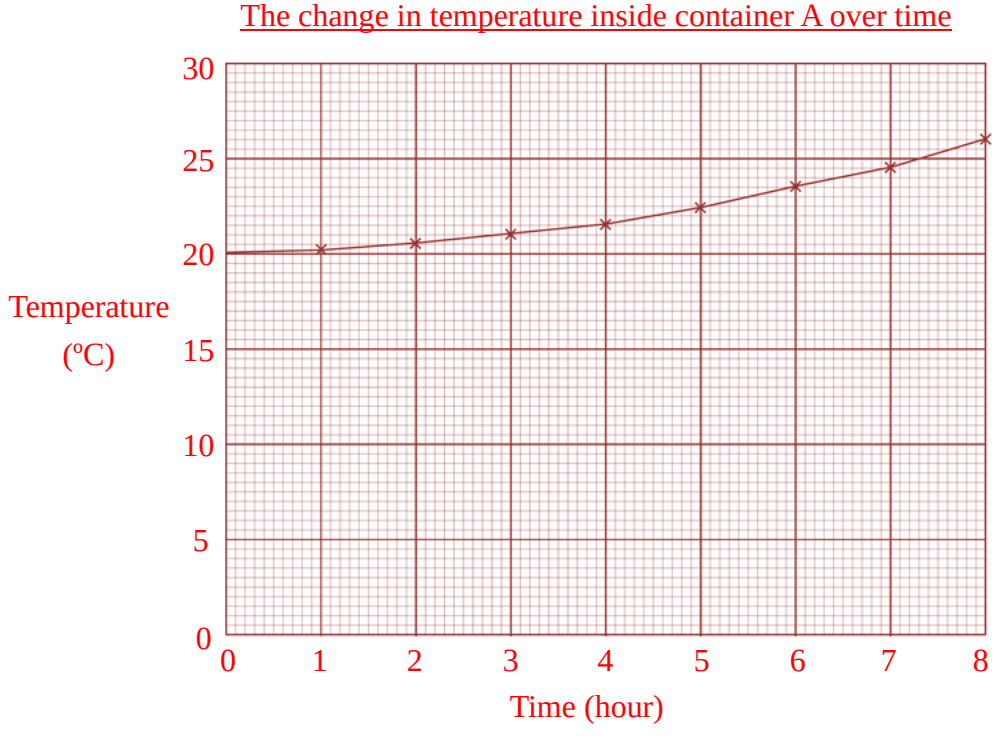
(c) Predict the change in the amount of carbon dioxide in the air inside container A. (1 mark)

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(d) The results of the investigation would be different if the containers were not covered with cotton wool. State and explain this difference. (2 marks)

##

(a) Correct title (1)
Correct labelling of axes (1)
Correct plotting and joining of points (1)
e.g.



(b) The level of the coloured liquid in the right arm of the U-shaped capillary tube would rise. (1)
The seeds in container A would release heat during respiration. (1)
The resulting expansion of the air inside the test tube in container A would push down the coloured liquid in the left arm of the capillary tube. (1)
(c) The amount of carbon dioxide in the air would increase. (1)
(d) Without the cotton wool covering, the temperature increase in the containers would be less than before/ the temperature in the containers would not increase. (1)

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This is because part of the heat released by the seeds would be lost to the surroundings. (1)

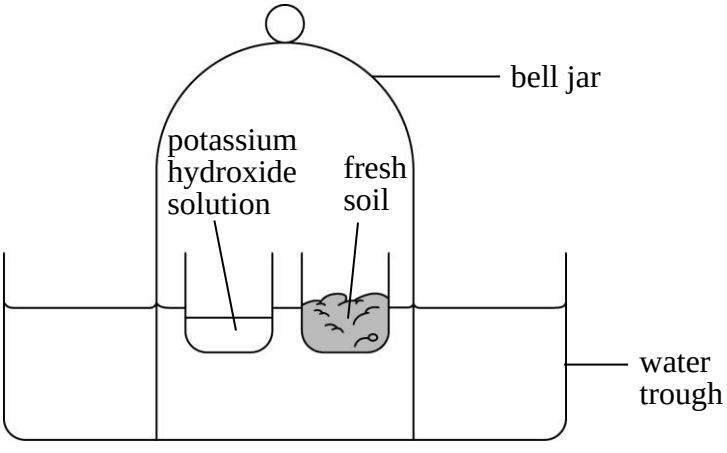
(9 marks)

##

	Skills assessed	Marks
(a)	Presenting information	3
(b)	Applying concepts	3
(c)	Applying concepts	1
(d)	Applying concepts	2

|!|BCELB0220000009|!| ★ **SI**

The following diagram shows an experiment to show the presence of living organisms in the soil.



(a) What is the function of the potassium hydroxide solution in the above set-up?

(1 mark)

(b) (i) How does the water level inside the bell jar change after a few hours?

(1 mark)

(ii) Account for the change in (i). (4 marks)

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(c) How can you set up a control for this experiment? (1 mark)

##

(a) It absorbs the carbon dioxide inside the bell jar. (1)

(b) (i) The water level inside the bell jar rises slightly. (1)

(ii) Living organisms in the fresh soil carry out respiration. (1)

They absorbed oxygen from the air inside the bell jar and (1)

released carbon dioxide, which was absorbed by the potassium hydroxide solution. (1)

Thus the air pressure in the bell jar decreased below that of the atmosphere, which pushes water in the trough up the bell jar. (1)

(c) A similar set-up can be used but with fresh soil replaced by baked soil. (1)

(7 marks)

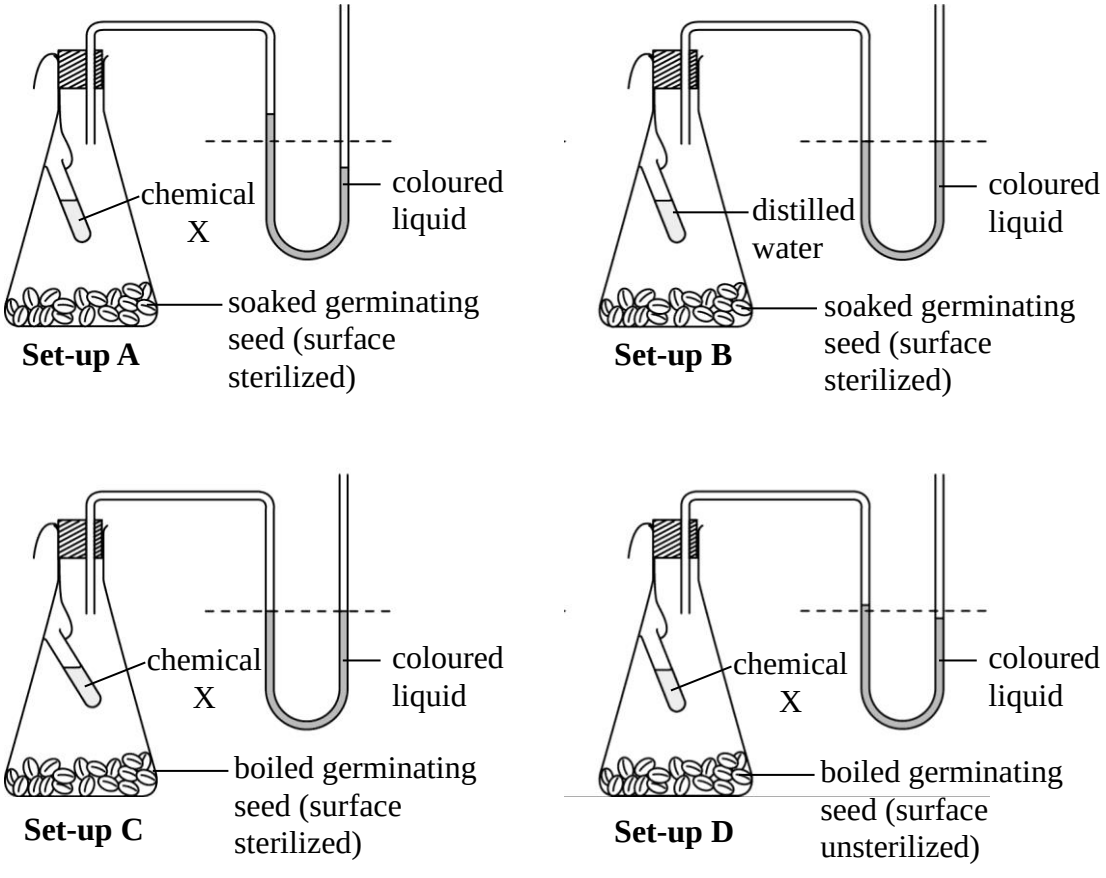
##

	Skills assessed	Marks
(a)	Practical skills	1
(b)(i)	Applying concepts	1
(b)(ii)	Applying concepts	4
(c)	Practical skills	1

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|!|BCELB0220000010|!| ★ **SI**

A student prepared four set-ups (A to D) to investigate the respiration of germinating seeds. The four set-ups were kept under identical conditions. The diagrams below show their appearances after three days. The dotted lines indicate the initial levels of the coloured liquid.



(a) Name chemical X and state its function. (2 marks)

_____

_____

(b) Which set-up(s) serve(s) as the control(s) for set-up A? (1 mark)

_____

(c) Explain the results of set-up A. (4 marks)

_____

_____

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(d) Why did the level of coloured liquid remain unchanged in set-up C? (1 mark)

##

(a) Potassium hydroxide solution (1)
It absorbs carbon dioxide. (1)
(b) Set-ups B and C (1)
(c) The seeds took in oxygen and released carbon dioxide. (1)
The carbon dioxide was absorbed by chemical X / the potassium hydroxide solution. (1)
The air pressure inside the flask decreased and became lower than the atmospheric pressure, (1)
so the coloured liquid was sucked towards the flask. (1)
(d) Both the seeds and microorganisms on the seed surface were killed and no respiration took place. (1)

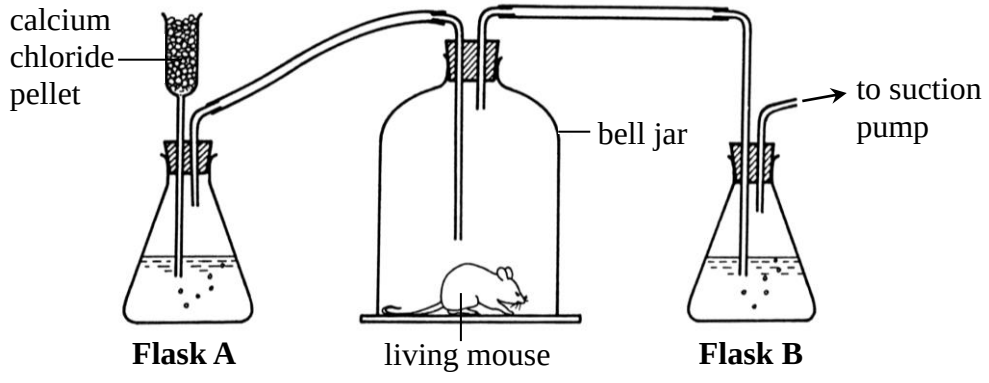
(8 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Practical skills	1
(c)	Applying concepts	4
(d)	Applying concepts	1

|!|BCELB0220000011|!| ★ **SI**

The following diagram shows an experiment to investigate the respiration of a living mouse. A control was also set up by using a similar set-up but with the mouse removed from the bell jar.



(a) The set-up above cannot give conclusive results. Suggest and explain how you would correct it. (2 marks)

(b) What solution should be used in flasks A and B? (1 mark)

(c) What is the purpose of the solution in flasks A and B, respectively? (2 marks)

(d) After your correction in (a), how would the colours of the solutions in flasks A and B change by the end of this experiment, respectively? (2 marks)

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(e) The mouse is replaced by a potted plant in order to investigate the respiration of the plant. Suggest, with reason(s), ***two*** other modifications to the set-up. (4 marks)

##

(a) Replace the calcium chloride pellets with potassium hydroxide pellets (1)
so as to absorb carbon dioxide from the incoming air. (1)

(b) Lime water / hydrogencarbonate indicator solution (1)

(c) Flask A: to check whether the incoming air is free of carbon dioxide (1)
Flask B: to show that carbon dioxide is coming from the mouse in the bell jar (1)

(d) In flask A, the lime water would remain clear **OR** hydrogencarbonate indicator solution would remain red. (1)
In flask B, the lime water would turn milky **OR** hydrogencarbonate indicator solution would turn yellow. (1)

(e) Cover the bell jar with black cloth / aluminium foil. (1)
This is because the plant can carry out photosynthesis and absorbs carbon dioxide in the presence of light. (1)
Wrap the pot with a plastic bag. (1)
This is because the microorganisms in the soil release carbon dioxide, which may interfere with the results. (1)

(11 marks)

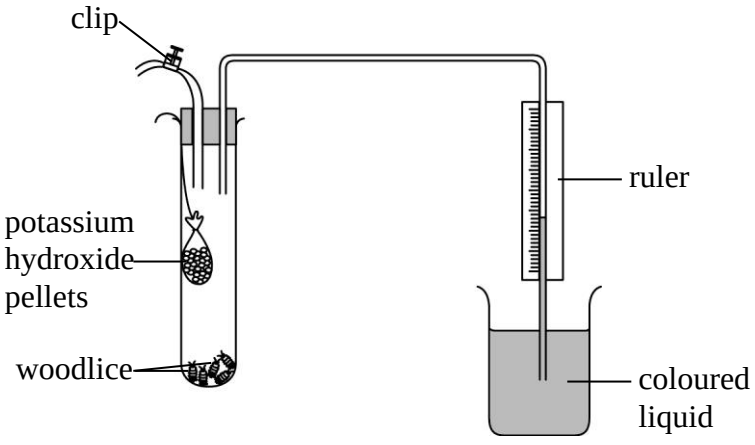
##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Practical skills	1
(c)	Practical skills	2
(d)	Applying concepts	2
(e)	Practical skills	4

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|!|BCELB0220000012|!| ★ **SI**

The following diagram shows an experimental set-up used to investigate the respiration of woodlice.



(a) What is the function of the potassium hydroxide pellets? (1 mark)

(b) How would you set up a control for this experiment? (1 mark)

(c) The following table shows the readings of the coloured liquid in both experimental and control set-ups.

	Experimental set-up	**Control set-up**
At 9:00 a.m.	30 mm^3	30 mm^3
At 12:00 p.m.	65 mm^3	20 mm^3

Calculate the respiration rate of the woodlice in the set-up. Show your working. (2 marks)

(d) (i) Explain the increased reading in the experimental set-up. (4 marks)

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(ii) Explain the decreased reading in the control set-up. (2 marks)

##

(a) It absorbs carbon dioxide. (1)

(b) Use the same set-up without including the woodlice. (1)

(c) Respiration rate of the woodlice

$$= \frac{35 + 10}{3}$$ (1)

$= 15\ mm^3/\ hour$ (1)

(d) (i) Oxygen was used up by the woodlice for respiration. (1)

The carbon dioxide produced by respiration was absorbed by the potassium hydroxide pellets. (1)

The air pressure in the test tube decreased to a level that was lower than the atmospheric pressure. (1)

Coloured water was therefore sucked upwards towards the test tube. (1)

(ii) The air inside the test tube expanded (1)

due to the increase in temperature at noon. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Practical skills	1
(c)	Mathematical skills	2
(d)(i)	Applying concepts	4
(d)(ii)	Applying concepts	2

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|!|BCELA0220000013|!| ★★ **Cross topic**

During a 100 m dash, a sprinter's muscles have a high demand for oxygen that may be beyond the circulatory system's capacity to fulfil. Under such conditions, both aerobic and anaerobic respiration might occur simultaneously.

(a) Describe how muscle cells obtain oxygen from the lungs. (4 marks)

(b) Outline the process of anaerobic respiration in muscle cells. (3 marks)

(c) The speed of a sprinter decreases during a race. Suggest ***two*** reasons why. (2 marks)

(d) Sprinters often consume glucose-rich drinks just before a race. In contrast, marathon athletes may include 'carbo-loading' in their training, during which they eat large amounts of starch-rich food. Explain why

(i) glucose-rich drinks are useful to sprinters. (1 mark)

(ii) 'carbo-loading' is useful to marathon athletes. (1 mark)

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##

(a) In the air sacs of the lungs, oxygen diffuses across the walls of the air sacs and the capillaries into the blood. (1)
Oxygen then binds with haemoglobin to form oxyhaemoglobin. (1)
Oxhaemoglobin are transported by the blood to the muscles, where the oxygen concentration is low. (1)
Oxyhaemoglobin releases its bound oxygen, which diffuses into the muscle cell. (1)

(b) Glucose is broken down through glycolysis to produce pyruvate and ATP. (1)
The pyruvate is then reduced by NADH (1)
to form lactic acid. (1)

(c) The build-up of lactic acid causes muscle fatigue. (1)
The amount of energy released from anaerobic respiration is low. (1)

(d) (i) Glucose can be utilized to release energy rapidly. (1)
(ii) Energy can be released from stored glycogen slowly. (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Understanding basic concepts	3
(c)	Applying concepts	2
(d)(i)	Applying concepts	1
(d)(ii)	Applying concepts	1

|!|BCELA0220000014|!| ★★ **Cross topic**

Beer-brewing uses certain species of yeast that can break down starch to form glucose, which is then fermented to make alcohol.

(a) Describe the formation of alcohol from glucose by yeast. (3 marks)

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(b) The brewing yeasts normally cannot digest all of the starch present and the beers they produce have comparatively high carbohydrate content. However, another species of yeast used in bread-making produces an enzyme that can digest relatively more starch. The brewing yeasts can be genetically modified to contain the gene that synthesizes this enzyme.

(i) Describe the procedure for producing these genetically modified (GM) yeasts. (4 marks)

(ii) Suggest how these GM yeast cells could produce beer that is low in carbohydrates. (2 marks)

##

(a) Glucose is first broken down to form pyruvate, ATP and NADH. (1)
In the absence of oxygen, the pyruvate then loses a carbon dioxide molecule (1)
and is reduced by NADH to produce alcohol in the form of ethanol. (1)

(b) (i) Obtain a DNA fragment containing the target gene (that codes for the desired enzyme) from the baking yeast. (1)
Cut a plasmid and the DNA fragment using the same restriction enzyme. (1)
Insert the target gene into the opened plasmid and join them with a DNA ligase. (1)
Introduce the recombinant plasmid into a brewing yeast. (1)

(ii) The yeast produces the enzyme that digests more starch into glucose. (1)
Glucose is further fermented, so less starch and glucose remain in the beer. (1)

(9 marks)

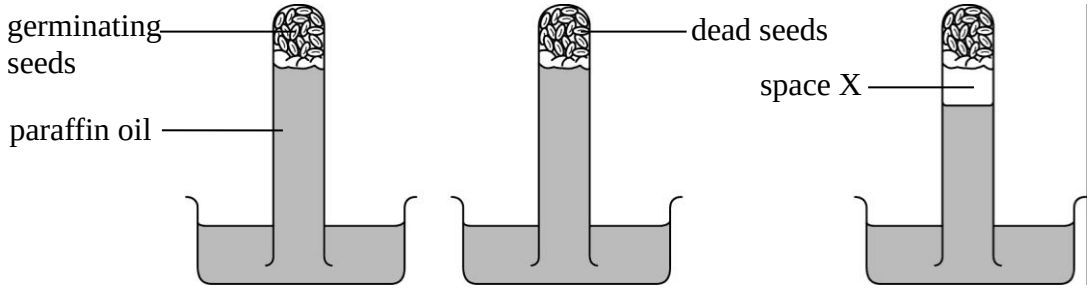
##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)(i)	Applying concepts	4
(b)(ii)	Applying concepts	2

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|!|BCELC0220000015|!| **SI**

The following diagram shows the set-ups (A and B) used to demonstrate the process of respiration in germinating seeds and the results of set-up A two days after the start of the experiment. The surfaces of the seeds were not sterilized.



(a) (i) What gas is present in space X? (1 mark)

(ii) How can you test for this gas? (3 marks)

(b) What would be observed in set-up B two days after the start of the experiment? Explain your answer. (3 marks)

##

(a) (i) Carbon dioxide (1)

(ii) Collect a gas sample in a syringe, (1)
then pump the gas into lime water / a hydrogencarbonate indicator solution. (1)
Lime water will turn milky / the hydrogencarbonate indicator will turn yellow. (1)

(b) The level of paraffin oil in the test tube would be slightly lower (1)
due to the presence of microorganisms / fungi on the surface of the dead seeds. (1)
These microorganisms carried out anaerobic respiration and thus released carbon dioxide. (1)

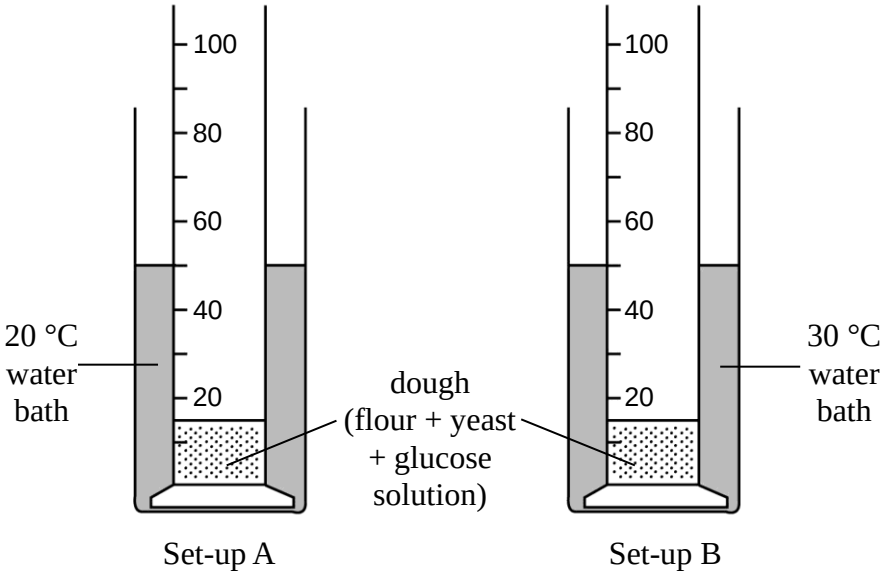
(7 marks)

##

	Skills assessed	Marks
(a)(i)	Applying concepts	1
(a)(ii)	Practical skills	3
(b)	Applying concepts	3

|!|BCELB0220000016|!| ★ **SI** **Cross topic**

The diagram below shows a set-up used to study anaerobic respiration in yeast at two different temperatures.



An increase in dough volume was observed in both set-ups A and B after an hour.

(a) Account for the increase in dough volumes. (2 marks)

(b) State and explain which set-up would show a greater increase in dough volume. (4 marks)

(c) Explain how the dough volume in set-up B would have been affected if the water bath had been set at 80 °C instead of 30 °C? (2 marks)

(d) Explain how the following variables of set-up B would have been affected if a greater amount of yeast had been used:

(i) the final dough volume reached (2 marks)

(ii) the rate of increase in dough volume (2 marks)

##

(a) Carbon dioxide was produced when the yeast respired anaerobically. (1)
The emissions caused the dough to rise. (1)

(b) Set-up B (1)
The higher temperature of the water bath in set-up B would lead to a higher level of enzyme activity in the yeast. (1)
Thus the rate of anaerobic respiration in the yeast of set-up B would be higher (1)
and more carbon dioxide would be produced within an hour. (1)

(c) The dough volume would have remained unchanged (1)
because the yeast would have been killed at 80 °C / the yeast enzymes would have been denatured at 80 °C. (1)

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(d) (i) The final dough volume reached would have remained unchanged because the amount of glucose (substrate) was fixed and the amount of carbon dioxide (product) produced by respiration would have been the same. (1) (1)

(ii) The rate of increase in dough volume would be higher because more carbon dioxide would be produced per unit time when more yeasts respired simultaneously. (1) (1)

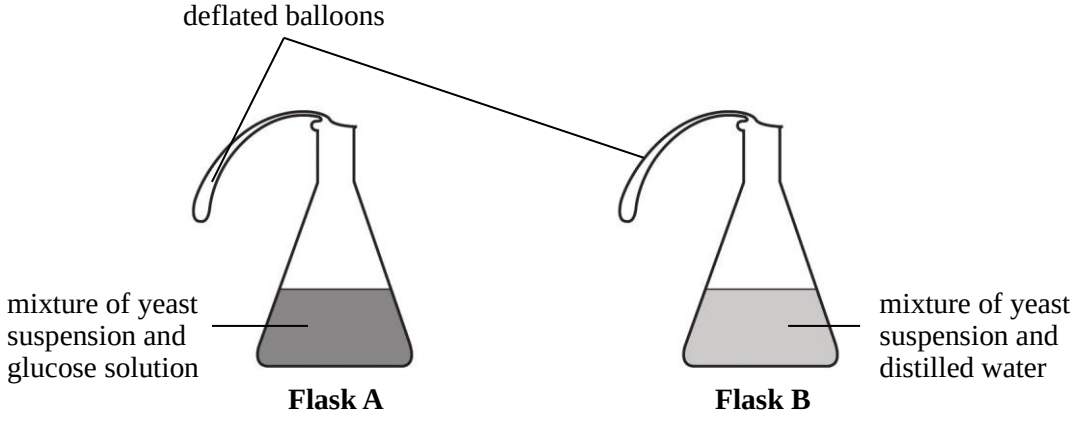
(12 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	4
(c)	Applying concepts	2
(d)(i)	Applying concepts	2
(d)(ii)	Applying concepts	2

|!|BCELB0220000017|!| ★ **SI** **Cross topic**

Sam set up two sealed flasks to investigate the anaerobic activity of yeast. The following diagram shows the set-ups.



(a) Suggest ***two*** factors that should be kept constant in this investigation. (2 marks)

(b) (i) After two days, there was a change in the size of the balloon attached to flask A. State and explain the change. (3 marks)

(ii) Based on the result from (b)(i), state ***one*** application of yeast in the food industry. (1 mark)

(c) Describe how you could measure the concentration of glucose in flask A as it changes during the two-day period. (4 marks)

##

(a) Any ***two*** of the following: (2)

- Temperature
- Concentration and volume of the yeast suspension
- Concentration and volume of the glucose solution

(Or other reasonable answers)

(b) (i) The balloon expanded (1)
because yeast breaks down glucose and evolves carbon dioxide (1)
when carrying out anaerobic respiration / alcoholic fermentation. (1)

(ii) Bread-making (1)

(c) Take samples from the reaction mixture at regular intervals. (1)
Heat the samples after they are put into test tubes with Benedict's solution. (1)
Measure the depth of the red precipitate that settles at the bottom of each test tube. (1)
Compare the measured data with a calibration curve (constructed using a glucose solution of known concentration) to determine the glucose concentration in the mixture. (1)

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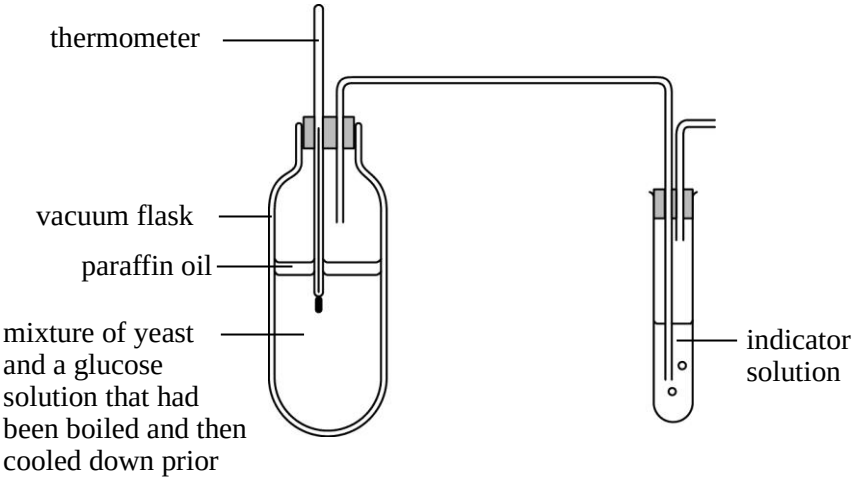
(10 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)(i)	Applying concepts	3
(b)(ii)	Understanding basic concepts	1
(c)	Practical skills	4

|!|BCELB0220000018|!| ★ **SI**

The diagram below shows the apparatus used in an experiment to investigate anaerobic respiration in yeast.



(a) Write a word equation for anaerobic respiration in yeast. (2 marks)

(b) Why was the glucose solution boiled and then cooled down beforehand? (3 marks)

(c) Explain how the temperature in the vacuum flask would change during the experiment. (2 marks)

(d) The indicator solution in the test tube was used to test for the product of anaerobic respiration in yeast.

(i) Name the product. (1 mark)

(ii) Give the name of the indicator solution and describe its colour change when the product is present. (2 marks)

##

(a) glucose → 2 ethanol + 2 carbon dioxide + energy (2 ATP) (2 or 0)

(b) Boiling the glucose solution removed any dissolved air that were present, (1)
which ensured that the respiration conducted by yeast cells would be anaerobic. (1)
The hot glucose solution had to be cooled down to avoid killing the added yeast. (1)

(c) The temperature would increase (1)
due to the heat released by yeast cells during anaerobic respiration. (1)

(d) (i) carbon dioxide (1)

(ii) Hydrogencarbonate indicator solution: (1)
turns from red to yellow (1)
OR
Lime water: (1)
turns from colourless to milky (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Practical skills	3
(c)	Applying concepts	2
(d)(i)	Understanding basic concepts	1
(d)(ii)	Practical skills	2

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|!|BCELA0220000019|!| ★★ **STSE**

Read the passage below and answer the questions that follow.

Low-alcohol beer

The market for low-alcohol beer in the United States is experiencing great growth due to the demand for healthy drinks, the increased concern over the social effects of excessive alcohol consumption and drunk-driving legislation. There are two approaches in producing low-alcohol beer: physical treatment of the beer and adaptation of the existing brewing process.

Physical treatment involves passing beer that is under high pressure through a reactor with a selectively permeable membrane. Alcohol and water molecules can pass through the membrane more readily than the larger molecules in the beer. Additional water is then added to the filtered beer to dilute it, thus producing the low-alcohol beer.

In comparison, adaptation of the existing brewing process can prevent too much alcohol from being produced in the first place. There are three methods to achieve this:

(a) Chill the yeast before fermentation
(b) Use fewer carbohydrates
(c) Create a new strain of yeast that converts fewer carbohydrates to alcohol

Some scientists believe that the third method is the most promising. They hope the new strain of yeast can be used to make a healthy and driver-friendly beer that does not differ greatly in taste.

(a) Why is high pressure applied to the beer in the physical treatment? (1 mark)

(b) Explain why the following methods can reduce the production of alcohol in yeast.

(i) Chilling the yeast before fermentation (2 marks)

(ii) Using fewer carbohydrates (2 marks)

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(c) Name the technology that can be used to create the new strain of yeast. (1 mark)

(d) State ***one*** example of the following benefits that low-alcohol beer can bring:

(i) social benefits (1 mark)

(ii) health benefits (1 mark)

##

(a) The high pressure increases the rate of movement of water and alcohol across the selectively permeable membrane. (1)

(b) (i) Alcoholic fermentation is catalysed by yeast enzymes. (1)
Low temperatures inactivate these enzymes and inhibit the fermentation. (1)

(ii) Carbohydrates are the substrate in alcoholic fermentation. (1)
The rate of reaction is limited by the concentration of substrates, so the rate of the fermentation can be lowered by reducing the amount of carbohydrates used. (1)

(c) Genetic engineering / recombinant DNA technology (1)

(d) (i) The number of traffic accidents caused by drunk-driving will be reduced. (1)
(Or other reasonable answers)

(ii) Liver damage caused by alcohol will be reduced. (1)
(Or other reasonable answers)

(8 marks)

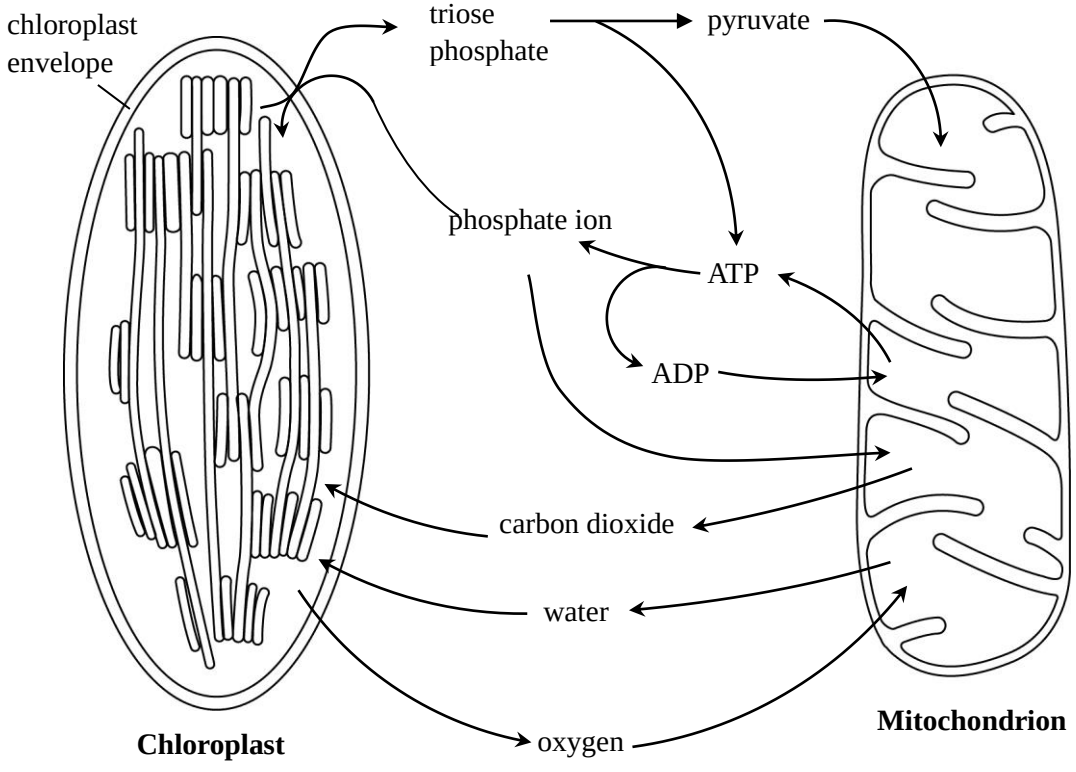
##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	2
(c)	Applying concepts	1
(d)(i)	Critical thinking	1
(d)(ii)	Critical thinking	1

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|!|BCELA0220000020|!| ★★ **Cross topic**

Palisade mesophyll cells have both chloroplasts and mitochondria. The following diagram shows the exchanges of substances between a mitochondrion, a chloroplast and the cytoplasm.



(a) Both the chloroplast and mitochondrion have unique internal membranous feature. Describe how these membranous features are related to the functions of the two organelles. (4 marks)

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(b) Explain why phosphate ions will not accumulate inside mitochondria. (2 marks)

(c) Triose phosphate leaves the chloroplast by passing through carrier proteins in the chloroplast envelope. These proteins also allow the passage of inorganic phosphate ion into the chloroplast at the same time.
Explain why the movement of triose phosphate out of the chloroplast is an example of diffusion rather than active transport. (2 marks)

(d) Both chloroplasts and mitochondria can synthesize ATP. Identify ***two*** differences between the processes of ATP synthesis in chloroplasts and mitochondria. (2 marks)

##

(a) The inner membrane of the mitochondrion is highly folded. (1)
This increases the surface area available for holding the enzymes and carriers that facilitate oxidative phosphorylation. (1)
The chloroplast contains a large number of thylakoids arranged in stacks called grana. (1)
This tight arrangement maximizes the amount of light energy that the chlorophyll on the thylakoid membranes captures for the production of NADPH and ATP. (1)

(b) Phosphate ions are used to produce ATP in oxidative phosphorylation / the Krebs cycle. (1)
The produced ATP then exits the mitochondrion. (1)

(c) Triose phosphate is the product of photosynthesis and so its concentration is higher in the chloroplast than in the cytoplasm. (1)

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Thus triose phosphate moves down its concentration gradient as it passes through the carrier protein without using energy. (1)

(d) Any ***two pairs*** of the following: (2)

	ATP synthesis in chloroplasts	**ATP synthesis in mitochondria**
Sources of energy	Sunlight trapped by chlorophyll	Chemical energy in food
Process	Electron in chlorophyll gets excited by sunlight. ATP is synthesized in photophosphorylation.	NADH / FADH from the Krebs cycle donates hydrogen (electron) to carriers in inner membrane of mitochondria. ATP is synthesized in oxidative phosphorylation.
Hydrogen (electron) donor	Water	NADH / FADH
Hydrogen (electron) acceptor	NADP	Oxygen

(10 marks)

##

	Skills assessed	**Marks**
(a)	Analysing data	4
(b)	Analysing data	2
(c)	Applying concepts	2
(d)	Applying concepts	2

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